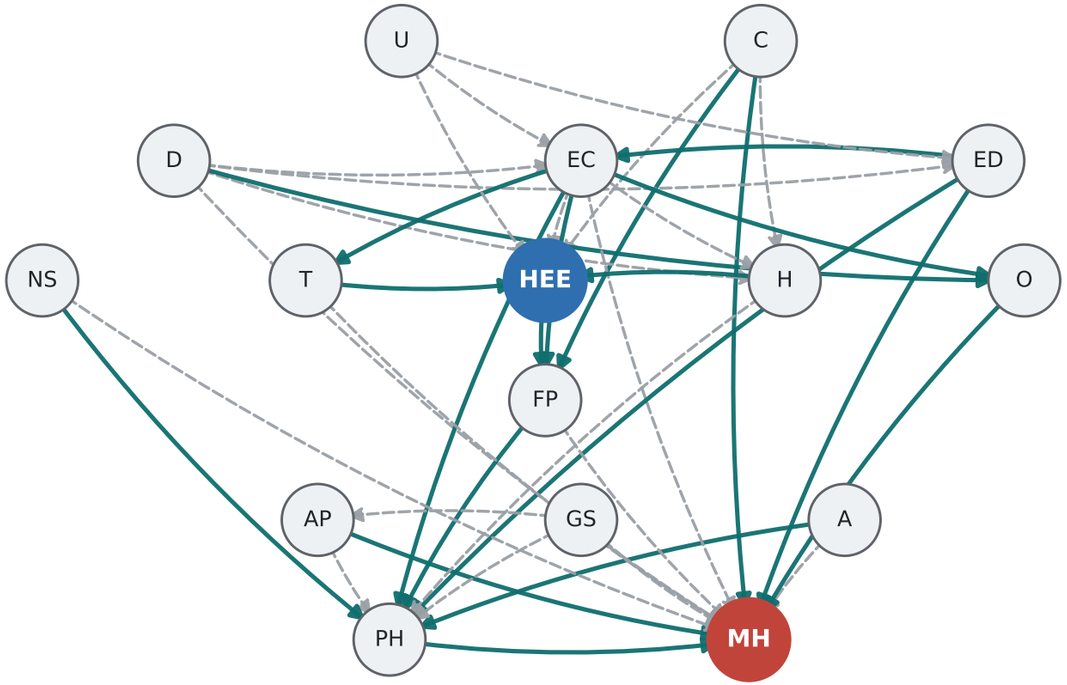
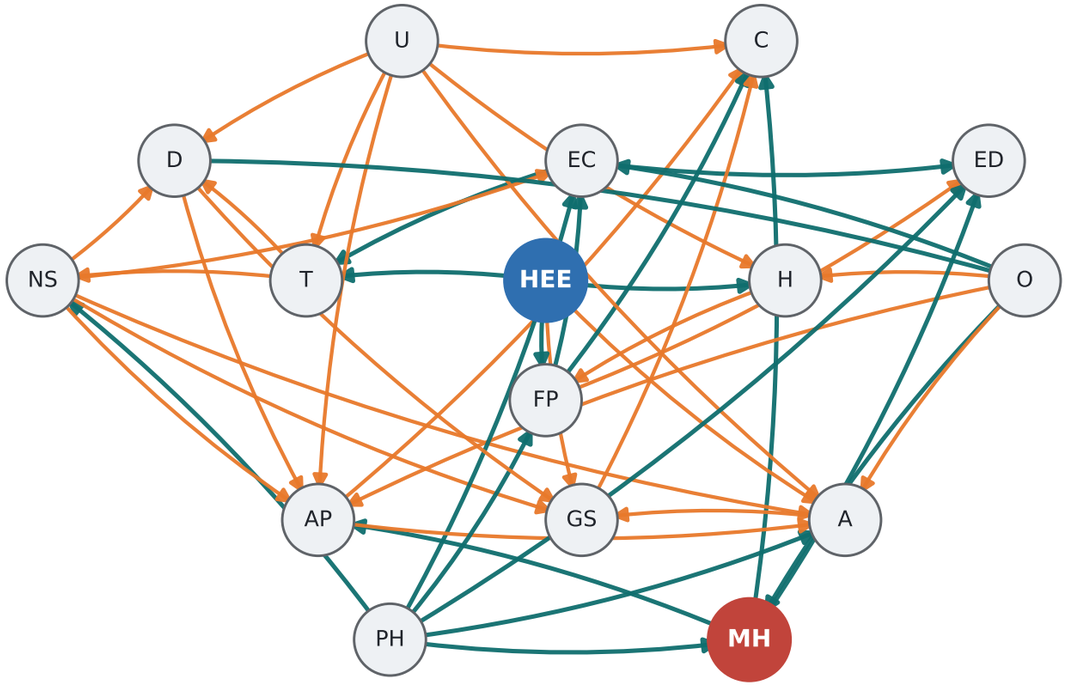
Expert-elicited DAG vs constraint-based causal discovery (HEE–MH structure, LSOA cross-section)

A. Expert DAG (induced subgraph, 16 measured constructs)



B. PC-stable discovery (data-driven, α = 0.01)



Shared node layout. PC recovers 19/40 expert adjacencies (skeleton recall 0.47); 26/45 PC edges are novel (precision 0.42). HEE and MH are non-adjacent in both graphs — consistent with the DAG's fully-mediated HEE→MH claim (no direct edge). Arrowheads on PC edges are CPDAG orientations (equivalence-class), plain lines are unoriented; all associations are area-level.

Shared adjacency (in both) — Expert-only (PC did not recover) — PC-only (novel / latent-projection) — HEE (exposure) — MH (outcome)

A = Accessibility AP = Air Pollution C = Climate D = Demographics EC = Economic Capital ED = Education FP = Fuel Poverty GS = Green Space
H = Housing Characteristics HEE = Home Energy Efficiency MH = Mental Health NS = Neighbourhood Stress O = Overcrowding PH = Physical Health T = Tenure U = Urbanicity